Name: _____

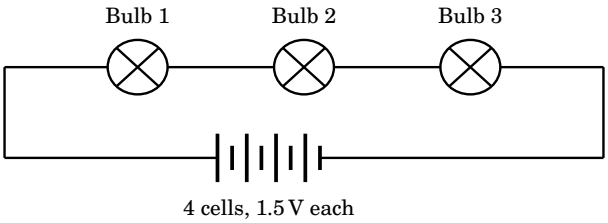
Date: _____

Mr. Rawson • Y9 MYP Physics

2.3 – Series Circuits: Build and Measure

Unit 2: Electricity & Energy

The question: *when three bulbs share one loop, how is the battery's voltage shared out, and what happens to the current when you take a bulb away?*



In a series circuit, the battery's voltage is SHARED between the bulbs.

Part 1 — Voltage in PhET

Open **Circuit Construction Kit: DC** (PhET).

1. Build a series circuit: **4 batteries and 3 bulbs** in one loop.
2. Click each battery and set its voltage to **1.5 V**.
3. Add a **voltmeter**. Put its two probes across **all 4 batteries**. Write the reading in the **PhET** column.
4. Move the probes across **Bulb 1**, then **Bulb 2**, then **Bulb 3**. Write each reading in the **PhET** column.

Part 2 — Voltage in your real circuit

5. Build the same circuit with real cells and bulbs.
6. Put the voltmeter probes across **all 4 cells**. Write the reading in the **Real** column.
7. Move the probes across **each bulb** in turn. Write each reading in the **Real** column.
8. Add the three bulb voltages together. Write the total in the last row.

Voltage across …	PhET (V)	Real circuit (V)
all 4 batteries (cells)	6.0	6.1
Bulb 1	2.0	2.0
Bulb 2	2.0	2.0
Bulb 3	2.0	1.9
Bulbs 1 + 2 + 3 added up	6.0	5.9

Check

In PhET, the three bulb voltages add up to the battery voltage **exactly**. In your real circuit they add up **close to** it. A small difference is normal, because the wires and the meter use up a little voltage themselves.

1) **Complete** the sentence.

The battery's voltage is _____ shared between the three bulbs, and the three bulb voltages _____ add up to the battery voltage.

Part 3 — Current: take a bulb out

9. Open the loop anywhere and put the **ammeter** in the gap, so the current goes *through* the meter. Read the current with **3 bulbs** and write down how bright the bulbs are.
10. **Predict** what the current will do when a bulb is taken out. Then take out **Bulb 3** and close the gap with a **lead**, so the loop is complete again. Read the ammeter and compare the brightness of the two bulbs left.
11. Take out **Bulb 2** and close the gap with a **lead**. Read the ammeter and compare the brightness of the last bulb.

Disconnect a lead after each reading.

Bulbs in the loop	My prediction (circle one)	Current (A)	Brightness
3	—	0.20	write how bright: dim
2	goes up / same / goes down	0.28	brighter / same / dimmer (brighter)
1	goes up / same / goes down	0.45	brighter / same / dimmer (brighter)

Brightness: in the last two rows, circle the word that compares the bulbs with the row above.

2) **Explain** why the current rises each time a bulb is taken out.

Model answer: *Each bulb makes it harder for the current to get round the loop. With fewer bulbs in the way, the same battery pushes more current round, so the bulbs that are left are brighter.*

Some students — the voltage rule and the brightness

The battery is **6 V** and the bulbs are identical. Use the voltage rule to find the voltage across **each** bulb with **3 bulbs**, with **2 bulbs** and with **1 bulb**. Then say what happens to the voltage across each bulb as bulbs are taken out.

3 bulbs: _____ 2 V 2 bulbs: _____ 3 V 1 bulb: _____ 6 V

Model answer: *The shared voltage is split between fewer bulbs, so each bulb gets a bigger share. More voltage across a bulb means more current through it, so it is brighter.*